Dr. Torsten Busch
Tel.: +49 (0) 69/257203 012
Fax: +49 (0) 69/257203 999
Email: torsten.busch@curiaglobal.com

Curia Germany GmbH
Process Research & Development
Industriepark Hoechst, D569
D-65926 Frankfurt am Main

July 23rd 2024

Graphene synthesis project update

1. Biochar synthesis

- Overall seven syntheses finished with hemp fibers from BAFA neu GmbH
  - three syntheses done with 0.02M sulfuric acid
    - first product not representative, not used further
    - second and third batch converted (see below)
  - four syntheses done with 0.10M sulfuric acid
    - very similar observations and outputs
    - products of these have been combined/homogenized
    - part of the combined lots has been converted

- Canadian Rockies Hemp arrived and cleared customs
  - several standard synthesis batches finished, seven of these (11.2 g) combined and homogenized
  - additional experiments done with higher concentrated sulfuric acid (1.5%, 2.0%, 5.0%)
  - one experiment performed under stirring, using cut up hemp fibers, product is finer and more homogeneous
  - one experiment performed with sulfuric acid pre-treatment of hemp (95 °C)

- Larger vessel reactivated, first production batch completed (14.9 g output)

2. Biochar analysis

- Sample of combined biochar (milled, MB2946X_gem) analyzed by SEM and Raman for reference

3. Graphene synthesis

- two experiments finished with steep heating ramp (20-30 °C/min)
  - biochar ground manually for first experiment (TB1131)
  - laboratory mill used for second experiment (TB1132)
  - products not yet characterized
  - no serious handling issues

- two experiments performed with much gentler heating ramp (3 °C/min, TB1133/1134)
  - better handling but no other apparent difference
- one experiment performed at 5 °C/min (MB2946)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h: minimal yield (MB2947A)
  - 800 °C, 2 h: minimal yield (MB2947B)
  - 800 °C, 3 h: no output (MB2947C)
  - ➔ Inertion apparently insufficient and material likely incinerated
  - 800 °C, 3 h repeat, much higher Ar stream: 38 mg output obtained (MB2947D)
  - 800 °C, 1 h, nitrogen inertion: 60 mg output (TB1135-1)
  - 800 °C, 1 h, increased nitrogen inertion: 70 mg output (TB1135-2)

- 2 g of combined biochar milled with 2 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 2 h, inertion equivalent to TB1135-2: 70 mg output (MB2952B)
  - 800 °C, 3 h, inertion equivalent to TB1135-2: 50 mg output (MB2952C)
    - ➔ smoldering observed upon contact of the cold material with air (see video)
  - 800 °C, 3 h, inertion equivalent to TB1135-2: 60 mg output (MB2952C2)
    - ➔ material wetted with water before/during removal from the apparatus

- Oxygen sensor shows complete inertion of apparatus within 3 minutes at selected nitrogen stream (equivalent to TB1135-2)

- 2 g of combined biochar milled with 3 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h, standard inertion: Quartz tube shattered, only 7 mg output (MB2955A)
  - 800 °C, 1 h, standard inertion (repeat of MB2955A): 90 mg output (MB2955A2)
  - 800 °C, 2 h, standard inertion: 80 mg output (TB1136)

- 2 g of combined biochar milled with 4 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h, standard inertion: 80 mg output (MB2962A)
  - 800 °C, 2 h, standard inertion: 90 mg output (MB2962B)

- 0.6 g of biochar (cut up, hydrolyzed under stirring) milled with 0.6 g of KOH and used for two batches on 0.2 – 0.3 g scale
  - 800 °C, 1 h, standard inertion: 70 mg output (MB2963A)
  - 800 °C, 2 h, standard inertion: 120 mg output (MB2963B)

- 2 g of combined biochar milled with 8 g of KOH and used for several batches on 0.25 g scale
  - 800 °C, 1 h, standard inertion: 60 mg output (TB1137-1)
  - 800 °C, 2 h, standard inertion: 60 mg output (TB1137-2)
  - 800 °C, 1 h, standard inertion: 70 mg output (TB1137-3)

- 2 g of biochar lot MB2959 milled with 12 g of KOH and used for several batches
  - 800 °C, 1 h, standard inertion: pending, but output low (≤ 20 mg, MB2965A)
  - 800 °C, 2 h, standard inertion: in progress (MB2965B)
  - 800 °C, 2 h, standard inertion, Oven B: in progress (MB2965B)

4. Graphene analysis

- SEM (and partially Raman) results available for
  - TB1133, MB2946A, MB2947D, TB1135-1, TB1135-2, MB2952B, MB2952C, MB2952C2, MB2955A, MB2955A2, TB1136, MB2962A/B, MB2963A/B, TB1137-1/2/3 (step 2 products)
  - 900561-500MG (Sigma-Aldrich reference)
  - 924458-500MG (Sigma-Aldrich reference "Single-layer graphene sheets for battery, Bio-sourced")

- NDA with TEM/BET provider signed, shipment of first sample in progress

5. Currently open questions

Experiment overview – Step 1

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output	
MB2928	AV1	6.0 g	BAFA neu Hemp Fibre VF	100.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.4 bar	742 g	Water	2.6 g	
Comment	Fibres not completely covered by liquid															
MB2933	AV1	6.1 g	BAFA neu Hemp Fibre VF	174.0 g	0.19%	0.02 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.5 bar	647 g	Water	1.2 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2935	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.9 g	0.19%	0.02 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	9.5 bar	500 g	Water	1.3 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2936	AV1	6.0 g	BAFA neu Hemp Fibre VF	172.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.8 bar	->	10.6 bar	589 g	Water	0.8 g	Combined and homogenized as lot no MRa231
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa225	AV1	6.0 g	BAFA neu Hemp Fibre VF	153.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	?	->	10.5 bar	500 g	Water	0.9 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa228	AV1	6.3 g	BAFA neu Hemp Fibre VF	154.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	11.0 bar	500 g	Water	0.9 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MRa231	AV1	6.1 g	BAFA neu Hemp Fibre VF	168.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.0 bar	->	10.5 bar	500 g	Water	0.8 g	
Comment	dark crusts on glassware, metal and liquid surface removed before filtration															
MB2937	AV1	9.4 g	Canadian Rockies Hemp	179.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	10.9 bar	569 g	Water	1.6 g	Combined and homogenzied as lot no MB2946X
Comment	very low amount of foreign material observed and removed during filtration															
MB2938	AV1	8.9 g	Canadian Rockies Hemp	175.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.2 bar	->	13.2 bar	577 g	Water	1.4 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2940	AV1	9.2 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	22.5 h	8.7 bar	->	10.6 bar	555 g	Water	1.5 g	
Comment	very low amount of foreign material observed and removed during filtration															
MB2942	AV1	8.9 g	Canadian Rockies Hemp	177.6 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h	8.3 bar	->	13.5 bar	578 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2943	AV1	9.8 g	Canadian Rockies Hemp	178.2 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.3 bar	->	12.7 bar	611 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2944	AV1	9.0 g	Canadian Rockies Hemp	178.4 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.4 bar	->	9.7 bar	593 g	Water	1.8 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2945	AV1	9.1 g	Canadian Rockies Hemp	177.7 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.7 bar	->	9.8 bar	6 g	Water	1.6 g	
Comment	low amount of foreign material observed and removed during filtration															
MB2948	AV1	9.4 g	Canadian Rockies Hemp	175.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	25 h	9.0 bar	->	12.6 bar	475 g	Water	1.7 g	
Comment	low amount of foreign material observed and removed during filtration															

Experiment	Reactor	Raw material		Acid				T	t	pressure			Wash		Output
MB2949	AV1	9.2 g	Canadian Rockies Hemp	174.3 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.4 bar	->	10.2 bar	538 g	Water	1.8 g
Comment	low amount of foreign material observed and removed during filtration														
MB2950	AV1	9.1 g	Canadian Rockies Hemp	170.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	23 h		->	11.1 bar	534 g	Water	1.6 g
Comment	low amount of foreign material observed and removed during filtration														
MB2951	AV1	9.3 g	Canadian Rockies Hemp	178.5 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	9.1 bar	->	11.2 bar	575 g	Water	1.8 g
Comment	low amount of foreign material observed and removed during filtration														
MRa255	AV1	9.2 g	Canadian Rockies Hemp	170.8 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	8.1 bar	->	12.8 bar	488 g	Water	1.7 g
Comment	some foreign material observed and removed during filtration														
MB2953	AV1	9.7 g	Canadian Rockies Hemp	177.1 g	0.96%	0.10 M	Sulfuric acid	180 °C	23.5 h	9.1 bar	->	10.4 bar	558 g	Water	1.8 g
Comment	low amount of foreign material observed and removed during filtration														
MB2956	AV1	9.8 g	Canadian Rockies Hemp	172.0 g	0.96%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	12.5 bar	548 g	Water	1.9 g
Comment	low amount of foreign material observed and removed during filtration														
MB2957	AV1	9.1 g	Canadian Rockies Hemp	176.8 g	1.50%	0.15 M	Sulfuric acid	180 °C	23.5 h	8.8 bar	->	8.5 bar	589 g	Water	1.5 g
Comment															
MB2958	AV1	9.3 g	Canadian Rockies Hemp	177.0 g	2.00%	0.20 M	Sulfuric acid	180 °C	24 h	8.9 bar	->	8.7 bar	562 g	Water	1.8 g
Comment															
MB2959	AV5	76.0 g	Canadian Rockies Hemp	953.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	25 h				3957 g	Water	14.9 g
Comment	low amount of foreign material observed and removed during filtration														
MB2960	AV1	3.8 g	Canadian Rockies Hemp (cut)	151.0 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.8 bar	->	10.5 bar	300 g	Water	0.6 g
Comment	Reaction performed with stirring														
MB2961	AV1	9.9 g	Canadian Rockies Hemp	172.0 g	5.00%	0.51 M	Sulfuric acid	180 °C	24 h	10.1 bar	->	16.1 bar	563 g	Water	1.7 g
Comment	low amount of foreign material observed and removed during filtration														
MB2964	AV1	9.0 g	Canadian Rockies Hemp	211.0 g + 140.8 g	1.00%	0.10 M	Sulfuric acid	180 °C	24 h	8.6 bar	->	8.8 bar	310 g + 603 g	Water	1.3 g
Comment	Hemp boiled with 1% H_2SO_4 at 95 °C, filtered and washed, then heated in Autoclave with fresh acid														

Experiment overview – Step 2

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
TB1131	A	1.2 g	MB2933	1.2 g	KOH	(90%)	manual	Ar	20-27 °C/min	790 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.16 g
Comment		ground biochar: brown powder, strong electrostatic charge										(+ water)		(atm. Pressure)		
TB1132	A	1.3 g	MB2935	1.3 g	KOH	(90%)	mill (2 min)	Ar	20-29 °C/min	792 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder										(+ water)		(atm. Pressure)		
TB1133	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	785 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1134	A	1.0 g	MRa231	1.0 g	KOH	(90%)	mill (2 min)	Ar	3 °C/min	771 °C	1 h	20 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2946	A	1.0 g	MB2946X	1.0 g	KOH	(90%)	mill (1 min)	Ar	5 °C/min	770 °C	1 h	22 g	10% HCl	100 °C	N2 stream	0.3 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2947A	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	803 °C	1 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	804 °C	2 h	10 g	10% HCl	40 °C	N2 stream	< 0.01 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		210 mbar		
MB2947C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	806 °C	3 h	Batch discarded				
Comment		ground biochar: brown powder; see temperature trend														
MB2947D	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	Ar	3 °C/min	810 °C	3 h	10 g	10% HCl	100 °C	N2 stream	38 mg
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-1	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1135-2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	14 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952B	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2952C	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	3 h	10 g	10% HCl	100 °C	N2 stream	0.05 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		

Experiment	Oven	Qty	Lot	Base			grinding	Gas	T (rate)	T (max)	t	Wash		Drying		Output
MB2952C2	A	0.25 g	MB2946X	0.25 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2955A	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	7 mg
Comment		Quartz tube shattered during experiment										(+ water)		(atm. Pressure)		
MB2955A2	A	0.25 g	MB2946X	0.38 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1136	A	0.26 g	MB2946X	0.39 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	14 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962A	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.08 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2962B	A	0.25 g	MB2946X	0.50 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.09 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963A	A	0.22 g	MB2960	0.22 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	10 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2963B	A	0.29 g	MB2960	0.29 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	10 g	10% HCl	100 °C	N2 stream	0.12 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-1	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-2	A	0.28 g	MB2946X	1.13 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	2 h	23 g	10% HCl	100 °C	N2 stream	0.06 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
TB1137-3	A	0.26 g	MB2946X	1.06 g	KOH	(90%)	mill (1 min)	N2	3 °C/min	800 °C	1 h	23 g	10% HCl	100 °C	N2 stream	0.07 g
Comment		ground biochar: brown powder; see temperature trend										(+ water)		(atm. Pressure)		
MB2965A	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	1 h	20 g	10% HCl	100 °C	N2 stream	
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend										(+ water)		(atm. Pressure)		
MB2965B	A	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h					
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend														
MB2965B2	B	0.22 g	MB2959	1.08 g	KOH	(90%)	mill (2.5 min)	N2	3 °C/min	800 °C	2 h					
Comment		ground biochar/KOH mixture likely absorbed some moisture; see temperature trend														